

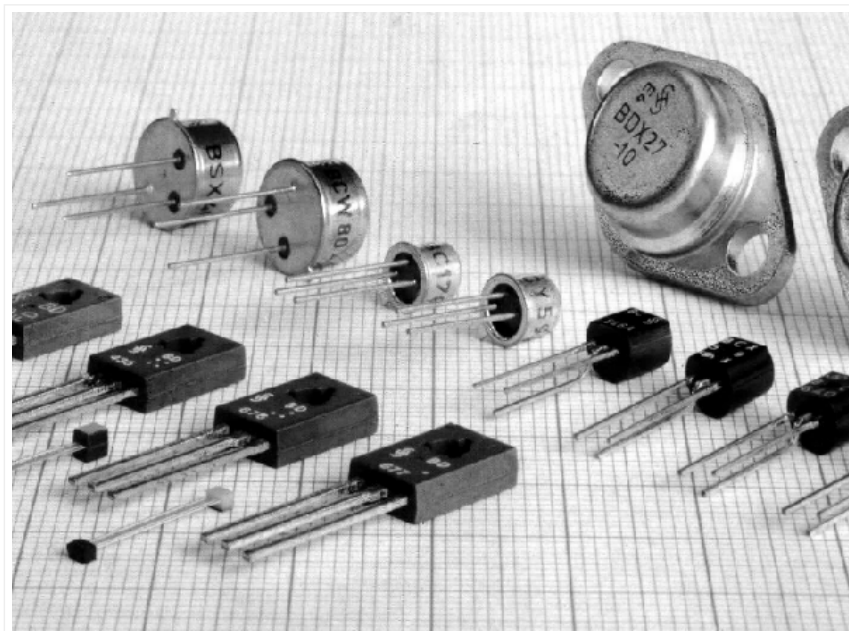
## 4. INTRODUCTION TO TRANSISTORS

# MIKROELEKTRONIKA

Transistors are active components and are found everywhere in electronic circuits. They are used as amplifiers and switching devices. As amplifiers, they are used in high and low frequency stages, oscillators, modulators, detectors and in any circuit needing to perform a function. In digital circuits they are used as switches.

There is a large number of manufacturers around the world who produce semiconductors (transistors are members of this family of components), so there are literally thousands of different types. There are low, medium and high power transistors, for working with high and low frequencies, for working with very high current and/or high voltages. Several different transistors are shown on 4.1.

The most common type of transistor is called bipolar and these are divided into NPN and PNP types. Their construction-material is most commonly silicon (their marking has the letter B) or germanium (their marking has the letter A). Original transistor were made from germanium, but they were very temperature-sensitive. Silicon transistors are much more temperature-tolerant and much cheaper to manufacture.



**Fig. 4.1: Different transistors**

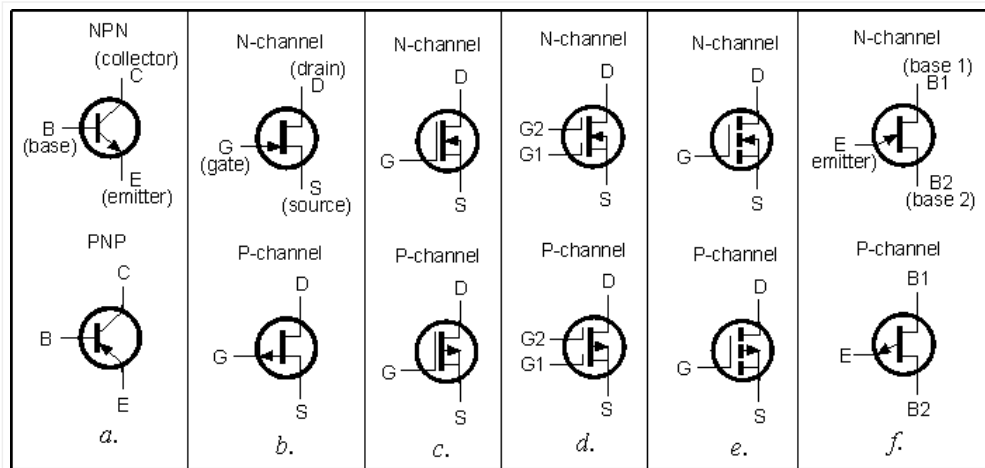


Fig. 4.2: Transistor symbols: a – bipolar, b – FET, c – **MOSFET**, d – **dual gate MOSFET**, e – **inductive channel MOSFET**, f – **single connection transistor**

The second letter in transistor's marking describes its primary use:

- C – low and medium power LF transistor,
- D – high power LF transistor,
- F – low power HF transistor,
- G – other transistors,
- L – high power HF transistors,
- P – photo transistor,
- S – switch transistor,
- U – high voltage transistor.

Here are few examples:

- AC540 – germanium core, LF, low power,
- AF125 – germanium core, HF, low power,
- BC107 – silicon, LF, low power (0.3W),
- BD675 – silicon, LF, high power (40W),
- BF199 – silicon, HF (to 550 MHz),
- BU208 – silicon (for voltages up to 700V),
- BSY54 – silicon, switching transistor.

There is a possibility of a third letter (R and Q – microwave transistors, or X – switch transistor), but these letters vary from manufacturer to manufacturer.

The number following the letter is of no importance to users.

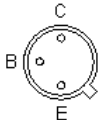
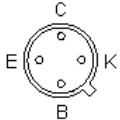
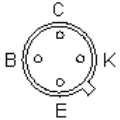
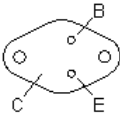
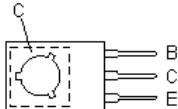
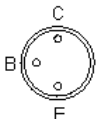
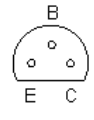
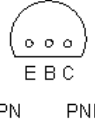
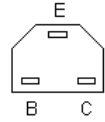
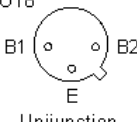
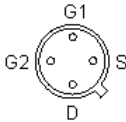
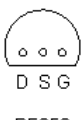
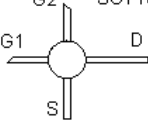
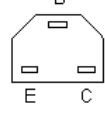
American transistor manufacturers have different marks, with a 2N prefix followed by a number (2N3055, for example). This mark is similar to diode marks, which have a 1N prefix (e.g. 1N4004).

Japanese bipolar transistor are prefixed with a: 2SA, 2SB, 2SC or 2SD, and FET-s with 3S:

- 2SA – PNP, HF transistors,
- 2SB – PNP, LF transistors,
- 2SC – NPN, HF transistors,
- 2SD – NPN, HF transistors.

Several different transistors are shown in photo 4.1, and symbols for schematics are on 4.2. Low power transistors are housed in a small plastic or metallic cases of various shapes. Bipolar transistors have three leads: for base (B), emitter (E), and for collector (C). Sometimes, HF transistors have another lead which is connected to the metal housing. This lead is connected to the ground of the circuit, to protect the transistor from possible external electrical interference. Four leads emerge from some other types, such as two-gate FETs. High power transistors are different from low-to-medium power, both in size and in shape.

It is important to have the manufacturer's catalog or a datasheet to know which lead is connected to what part of

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| <p>Package<br/>TO5, TO18</p>  <p>NPN      PNP</p> <p>BC107    BC177<br/>BC108    BC178<br/>BC109    BC179<br/>BC186<br/>BC187<br/>BC188<br/>BC189<br/>BC190<br/>BC191<br/>BC192<br/>BC193<br/>BC194<br/>BC195<br/>BC196<br/>BC197<br/>BC198<br/>BC199<br/>BC200</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <p>TO72</p> <div style="display: flex; justify-content: space-around;"> <div style="text-align: center;">  <p>NPN      PNP</p> <p>BF180    AF139<br/>BF181    AF178<br/>BF182    AF179<br/>BF183    AF180<br/>BF184    AF181<br/>BF200</p> </div> <div style="text-align: center;">  <p>NPN      PNP</p> <p>BF115    AF124<br/>BF116    AF125<br/>BF117    AF126<br/>BF118    AF127<br/>BF119<br/>BF120<br/>BF121<br/>BF122<br/>BF123<br/>BF124<br/>BF125<br/>BF126<br/>BF127<br/>BF128<br/>BF129<br/>BF130<br/>BF131<br/>BF132<br/>BF133<br/>BF134<br/>BF135<br/>BF136<br/>BF137<br/>BF138<br/>BF139<br/>BF140<br/>BF141<br/>BF142<br/>BF143<br/>BF144<br/>BF145<br/>BF146<br/>BF147<br/>BF148<br/>BF149<br/>BF150<br/>BF151<br/>BF152<br/>BF153<br/>BF154<br/>BF155<br/>BF156<br/>BF157<br/>BF158<br/>BF159<br/>BF160<br/>BF161<br/>BF162<br/>BF163<br/>BF164<br/>BF165<br/>BF166<br/>BF167<br/>BF168<br/>BF169<br/>BF170<br/>BF171<br/>BF172<br/>BF173<br/>BF174<br/>BF175<br/>BF176<br/>BF177<br/>BF178<br/>BF179<br/>BF180<br/>BF181<br/>BF182<br/>BF183<br/>BF184<br/>BF185<br/>BF186<br/>BF187<br/>BF188<br/>BF189<br/>BF190<br/>BF191<br/>BF192<br/>BF193<br/>BF194<br/>BF195<br/>BF196<br/>BF197<br/>BF198<br/>BF199<br/>BF200</p> </div> </div> | <p>TO3<br/>and similar</p>  <p>NPN      PNP</p> <p>2N3055    PNP3055<br/>BDY20    BDX18<br/>BDY21    OC26<br/>BDY22    AD149<br/>BDY23    AD162<br/>AD161</p>                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <p>TO126</p>  <p>NPN      PNP</p> <p>BD135    BD136<br/>BD137    BD138<br/>BD139    BD139<br/>BD140    BD140<br/>BD141    BD141<br/>BD142    BD142<br/>BD143    BD143<br/>BD144    BD144<br/>BD145    BD145<br/>BD146    BD146<br/>BD147    BD147<br/>BD148    BD148<br/>BD149    BD149<br/>BD150    BD150<br/>BD151    BD151<br/>BD152    BD152<br/>BD153    BD153<br/>BD154    BD154<br/>BD155    BD155<br/>BD156    BD156<br/>BD157    BD157<br/>BD158    BD158<br/>BD159    BD159<br/>BD160    BD160<br/>BD161    BD161<br/>BD162    BD162<br/>BD163    BD163<br/>BD164    BD164<br/>BD165    BD165<br/>BD166    BD166<br/>BD167    BD167<br/>BD168    BD168<br/>BD169    BD169<br/>BD170    BD170<br/>BD171    BD171<br/>BD172    BD172<br/>BD173    BD173<br/>BD174    BD174<br/>BD175    BD175<br/>BD176    BD176<br/>BD177    BD177<br/>BD178    BD178<br/>BD179    BD179<br/>BD180    BD180<br/>BD181    BD181<br/>BD182    BD182<br/>BD183    BD183<br/>BD184    BD184<br/>BD185    BD185<br/>BD186    BD186<br/>BD187    BD187<br/>BD188    BD188<br/>BD189    BD189<br/>BD190    BD190<br/>BD191    BD191<br/>BD192    BD192<br/>BD193    BD193<br/>BD194    BD194<br/>BD195    BD195<br/>BD196    BD196<br/>BD197    BD197<br/>BD198    BD198<br/>BD199    BD199<br/>BD200    BD200</p> | <p>TO-1</p>  <p>NPN      PNP</p> <p>AC176    AC128<br/>AC187    AC188</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <p>X-55</p>  <p>NPN      PNP</p> <p>BC182    BC212<br/>BC183    BC213<br/>BC184    BC214<br/>BC185    BC215<br/>BC186    BC216<br/>BC187    BC217<br/>BC188    BC218<br/>BC189    BC219<br/>BC190    BC220<br/>BC191    BC221<br/>BC192    BC222<br/>BC193    BC223<br/>BC194    BC224<br/>BC195    BC225<br/>BC196    BC226<br/>BC197    BC227<br/>BC198    BC228<br/>BC199    BC229<br/>BC200    BC230</p> | <p>TO92</p>  <p>NPN      PNP</p> <p>BC183L    BC213L<br/>BC237B    BC557<br/>BC547    BC307B<br/>BC548    2N4402<br/>BC549<br/>BC550<br/>BC551<br/>BC552<br/>BC553<br/>BC554<br/>BC555<br/>BC556<br/>BC557<br/>BC558<br/>BC559<br/>BC560<br/>BC561<br/>BC562<br/>BC563<br/>BC564<br/>BC565<br/>BC566<br/>BC567<br/>BC568<br/>BC569<br/>BC570<br/>BC571<br/>BC572<br/>BC573<br/>BC574<br/>BC575<br/>BC576<br/>BC577<br/>BC578<br/>BC579<br/>BC580<br/>BC581<br/>BC582<br/>BC583<br/>BC584<br/>BC585<br/>BC586<br/>BC587<br/>BC588<br/>BC589<br/>BC590<br/>BC591<br/>BC592<br/>BC593<br/>BC594<br/>BC595<br/>BC596<br/>BC597<br/>BC598<br/>BC599<br/>BC600</p> | <p>SOT25</p>  <p>NPN</p> <p>BF194<br/>BF195<br/>BF196<br/>BF197<br/>BF198<br/>BF199<br/>BF200</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <p>TO18</p>  <p>Unijunction<br/>transistor</p> <p>2N2646    2N4870<br/>2N2647    2N4871</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <p>G1</p>  <p>3N140<br/>3N141<br/>40673</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <p>D S G</p>  <p>BF256<br/>2N3819<br/>(connection FET)</p>                                                                                                                                                                                                                                                                                                                                                  | <p>G2 SOT103</p>  <p>BF960    BF981<br/>BF961    3SK81</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <p>B</p>  <p>NPN      PNP</p> <p>BC157    BC147<br/>BC158    BC148<br/>BC159    BC149<br/>BC160    BC150<br/>BC161    BC151<br/>BC162    BC152<br/>BC163    BC153<br/>BC164    BC154<br/>BC165    BC155<br/>BC166    BC156<br/>BC167    BC157<br/>BC168    BC158<br/>BC169    BC159<br/>BC170    BC160<br/>BC171    BC161<br/>BC172    BC162<br/>BC173    BC163<br/>BC174    BC164<br/>BC175    BC165<br/>BC176    BC166<br/>BC177    BC167<br/>BC178    BC168<br/>BC179    BC169<br/>BC180    BC170<br/>BC181    BC171<br/>BC182    BC172<br/>BC183    BC173<br/>BC184    BC174<br/>BC185    BC175<br/>BC186    BC176<br/>BC187    BC177<br/>BC188    BC178<br/>BC189    BC179<br/>BC190    BC180<br/>BC191    BC181<br/>BC192    BC182<br/>BC193    BC183<br/>BC194    BC184<br/>BC195    BC185<br/>BC196    BC186<br/>BC197    BC187<br/>BC198    BC188<br/>BC199    BC189<br/>BC200    BC19</p> |

It might be useful to remember the pinout for TO-1, TO-5, TO-18 and TO-72 packages and compare them with the drawing 4.2 (a). These transistors are the ones you will come across frequently in everyday work.

Transistors used with very high frequencies (like BFR14) have pins shaped differently.

As we said, there are literally thousands of different transistors, many of them have similar characteristics, which makes it possible to replace a faulty transistor with a different one. The characteristics and similarities can be

found in comparison charts. If you do not have one these charts, you can try some of the transistors you already have. If the circuit continues to operate correctly, everything is ok. You can only replace an NPN transistor with an NPN transistor. The same goes if the transistor is PNP or a FET. It is also necessary to make sure the pinout is correct, before you solder it in place and power up the project.

As a helpful guide, there is a chart in this chapter which shows a list of replacements for some frequently used transistors.



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